

DUSHANBE, TajikistanWMO: **388360**Lat: **38.583N**Lon: **68.733E**Elev: **2625**StdP: **13.35**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	21.4	8.3	9.3	28.4	12.2	11.3	33.7	16.6	47.8	15.0	48.8	2.4	270	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.0	100.8	66.9	98.7	65.9	96.7	65.4	72.5	92.7	70.2	90.9	68.4	89.2	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	105.4	83.9	63.2	95.8	80.8	61.1	88.6	77.8	38.0	92.8	35.9	90.7	34.3	89.8	92.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.5	10.7	11.2	105.0	7.1	1.9	6.1	106.4	2.0	107.5	-2.0	108.5	-7.1	109.9
			10.9	75.6	7.0	2.2	5.8	77.2	1.7	78.5	-2.2	79.8	-7.3	81.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	38.9	41.6	52.8	60.8	69.0	77.4	81.5	79.2	71.3	61.0	50.2	40.7
	DBStd	16.18	7.84	9.10	7.78	7.30	5.41	4.44	3.40	3.57	4.55	6.14	7.50	7.44
	HDD50	1051	346	251	61	11	0	0	0	0	0	4	83	295
	HDD65	3386	808	655	380	159	24	0	0	0	8	153	446	753
	CDD50	4876	3	16	149	336	589	822	977	905	639	344	89	7
	CDD65	1736	0	0	3	34	148	372	512	440	197	28	2	0
	CDH74	24536	0	3	49	483	1776	4960	7565	6217	2842	596	45	0
	CDH80	13356	0	0	4	118	687	2730	4606	3686	1349	169	7	0
	WSAvg	3.6	3.4	4.3	4.9	4.1	3.9	4.4	3.7	3.2	3.0	2.7	2.6	3.2
	PrecAvg	26.4	3.1	3.6	5.3	4.5	3.2	0.3	0.1	0.0	0.1	1.5	2.1	2.8
(15) Precipitation	PrecMax	36.7	6.3	7.0	9.8	8.8	8.0	1.6	1.5	0.2	1.4	6.7	5.0	7.7
	PrecMin	16.2	0.4	0.5	1.8	1.5	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.4
	PrecStd	5.3	1.3	1.8	1.9	1.8	2.0	0.4	0.3	0.0	0.3	1.6	1.5	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	69.6	79.2	87.9	94.1	100.8	105.1	103.6	96.4	89.4	80.4	66.0
		MCWB	46.5	49.5	55.8	62.5	64.2	66.4	68.5	67.6	61.6	57.9	54.5	48.2
	2%	DB	58.6	62.4	74.8	83.3	90.0	98.5	102.3	100.2	93.5	84.4	72.4	60.7
		MCWB	44.8	48.7	55.2	60.7	63.9	65.2	68.5	66.1	61.2	57.6	53.0	45.6
	5%	DB	54.2	58.1	70.3	79.8	87.4	96.1	100.1	97.9	91.4	80.7	68.0	56.8
		MCWB	43.2	45.7	53.0	60.0	63.3	65.1	66.8	65.0	61.2	57.1	51.1	44.4
	10%	DB	50.3	55.0	66.4	75.6	84.0	93.3	97.1	95.2	88.0	76.7	63.6	52.6
		MCWB	41.7	44.3	51.3	58.1	62.7	64.6	66.4	64.1	60.4	55.8	49.4	42.7
	0.4%	WB	49.3	52.7	59.5	65.9	68.6	72.5	78.4	73.7	67.9	62.1	57.3	50.5
		MCDB	58.5	64.4	74.5	80.3	85.6	93.0	96.7	96.0	89.8	80.2	74.3	60.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.6	49.5	56.6	63.3	66.5	70.0	74.2	70.7	64.9	59.7	54.6	47.7
		MCDB	53.7	58.8	70.2	78.5	83.4	89.8	92.0	92.0	86.6	77.9	68.6	55.8
	5%	WB	44.7	47.6	54.5	61.5	65.0	68.2	71.4	68.5	62.7	58.1	52.5	46.0
		MCDB	51.4	55.4	66.3	75.5	81.8	88.0	92.2	90.1	84.5	76.4	64.1	52.7
	10%	WB	42.9	45.7	52.6	59.8	63.7	66.6	69.2	66.6	61.2	56.5	50.7	44.1
		MCDB	48.8	52.3	63.3	72.2	79.7	86.8	90.6	88.9	83.6	73.5	60.7	50.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	17.3	19.5	21.9	24.7	28.5	31.0	32.2	32.2	28.4	21.6	18.5
		MCDBR	23.0	23.0	25.9	29.6	31.5	33.0	35.1	35.3	36.7	34.3	30.2	25.7
		MCWBR	12.8	12.5	12.8	13.3	12.4	11.9	12.2	12.1	13.4	13.7	14.1	13.9
		MCDBR	18.8	18.9	22.0	24.8	26.7	29.4	30.7	32.3	33.2	29.6	24.5	19.9
	5% WB	MCWBR	11.3	11.2	11.8	12.5	11.7	12.1	13.1	13.0	13.2	12.8	12.5	12.0
(40) Clear-Sky Solar Irradiance	taub		0.369	0.421	0.492	0.505	0.517	0.547	0.546	0.504	0.464	0.485	0.432	0.374
	taud		2.098	1.964	1.839	1.866	1.880	1.814	1.811	1.898	1.955	1.865	1.960	2.067
	Ebn at Noon		244	246	240	247	247	239	238	245	246	220	216	230
	Edh at Noon		37	48	61	62	63	67	67	60	53	53	42	36
(44) All-Sky Solar Radiation	RadAvg		650	846	1199	1627	2020	2384	2433	2259	1870	1261	756	578
	RadStd		83	138	120	182	159	115	78	88	67	122	102	79

Historical Trends

		Heating			Cooling		
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